

Taller 03: series de Taylor y polinomios de Lagrange

Grafique las curvas de las series de Taylor de varios órdenes para los siguientes casos:=0

```
import sympy as sp

x = sp.Symbol('x')
f = sp.cos(x)

# Orden
ordenes = [0, 1, 2, 3, 4]

# Funcion
graficas = [f]
labels = ['cos(x)']

for i in range(5):
    polinomio = sp.series(f, x, 0, n=i+1).removeO()
    print(f"Orden {i}: {polinomio}")

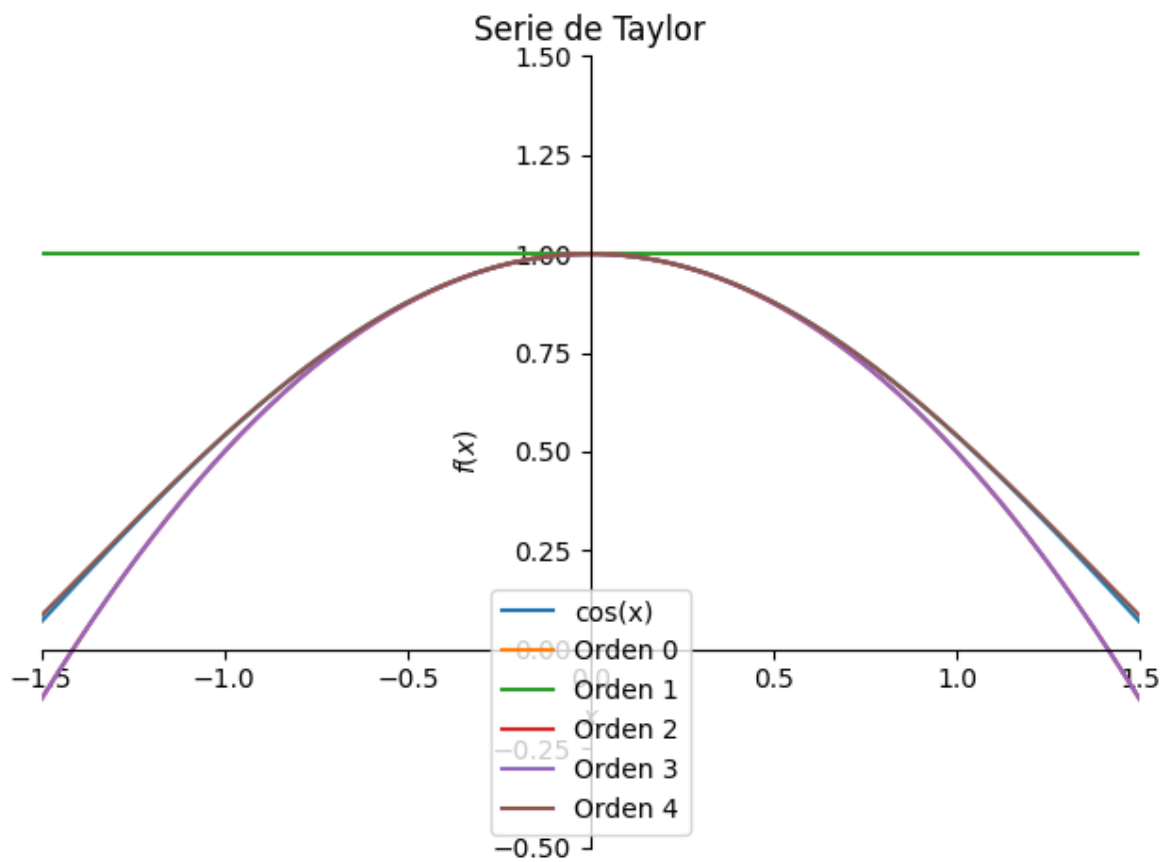
for n in ordenes:
    p = sp.series(f, x, 0, n+1).removeO()
    graficas.append(p)
    labels.append(f'Orden {n}')

p_plot = sp.plot(*graficas, (x, -1.5, 1.5),
                  xlim=(-1.5, 1.5),
                  ylim=(-0.5, 1.5),
                  show=False,
                  title='Serie de Taylor',
                  legend=True)
```

```
# Leyenda
for i, label in enumerate(labels):
    p_plot[i].label = label

p_plot.show()
```

```
Orden 0: 1
Orden 1: 1
Orden 2: 1 - x**2/2
Orden 3: 1 - x**2/2
Orden 4: x**4/24 - x**2/2 + 1
```



```
import sympy as sp

x = sp.Symbol('x')
```

```

f = sp.cos(x)

# Orden
ordenes = [0, 1, 2, 3, 4]

# Funcion
graficas = [f]
labels = ['cos(x)']

for i in range(5):
    polinomio = sp.series(f, x, 1, n=i+1).removeO()
    print(f"Orden {i}: {polinomio}")

for n in ordenes:
    p = sp.series(f, x, 1, n+1).removeO()
    graficas.append(p)
    labels.append(f'Orden {n}')

p_plot = sp.plot(*graficas, (x, -1.5, 1.5),
                 xlim=(-1.5, 1.5),
                 ylim=(-0.5, 1.5),
                 show=False,
                 title='Serie de Taylor',
                 legend=True)

# Etiqueta
for i, label in enumerate(labels):
    p_plot[i].label = label

p_plot.show()

```

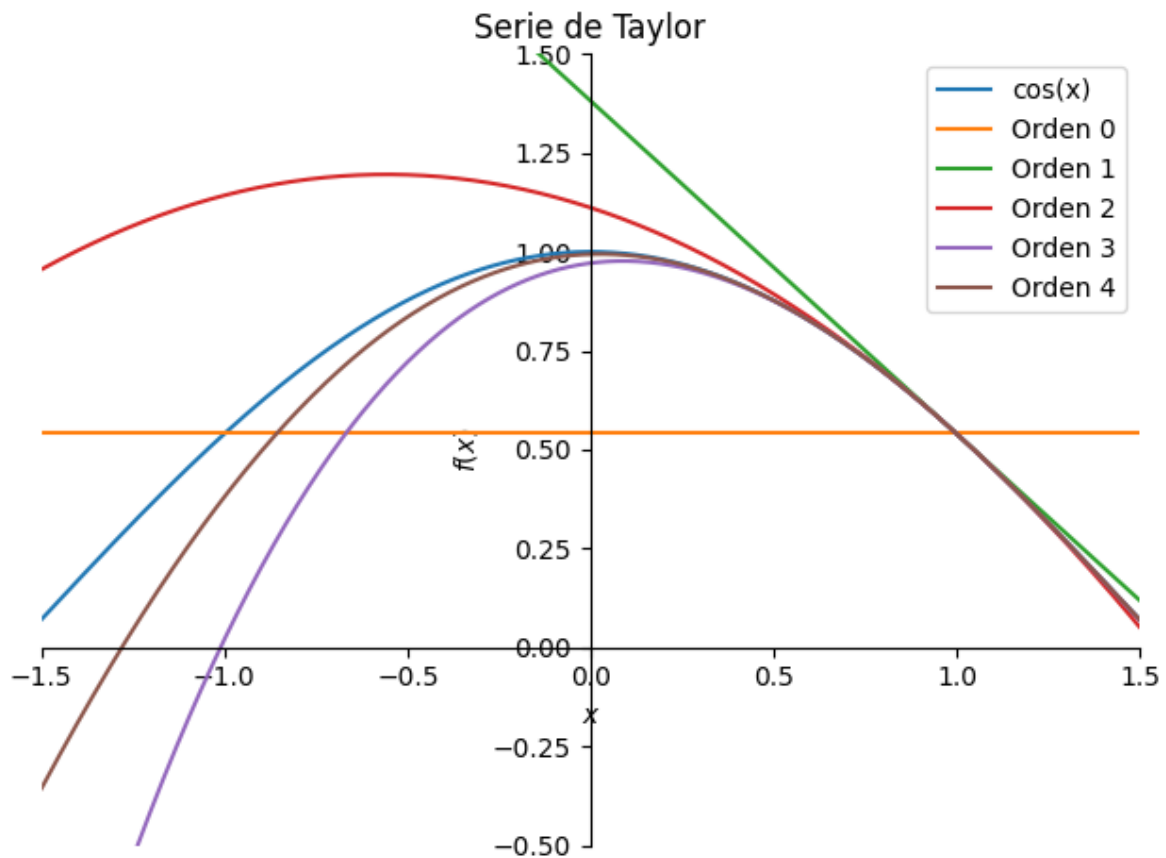
Orden 0: $\cos(1)$

Orden 1: $-(x - 1)\sin(1) + \cos(1)$

Orden 2: $-(x - 1)^2\cos(1)/2 - (x - 1)\sin(1) + \cos(1)$

Orden 3: $(x - 1)^3\sin(1)/6 - (x - 1)^2\cos(1)/2 - (x - 1)\sin(1) + \cos(1)$

Orden 4: $(x - 1)^4\cos(1)/24 + (x - 1)^3\sin(1)/6 - (x - 1)^2\cos(1)/2 - (x - 1)\sin(1) +$



```
import sympy as sp

x = sp.Symbol('x')
f = 1/(1-x)

# Orden
ordenes = [0, 1, 2, 3, 4]

# Funcion
graficas = [f]
labels = ['1/(1-x)']

for i in range(5):
    polinomio = sp.series(f, x, 0, n=i+1).removeO()
    print(f"Orden {i}: {polinomio}")

for n in ordenes:
```

```

p = sp.series(f, x, 0, n+1).remove0()
graficas.append(p)
labels.append(f'Orden {n}')

p_plot = sp.plot(*graficas, (x, -1.5, 1.5),
                  xlim=(-1.5, 1.5),
                  ylim=(-0.5, 1.5),
                  show=False,
                  title='Serie de Taylor',
                  legend=True)

# Etiqueta
for i, label in enumerate(labels):
    p_plot[i].label = label

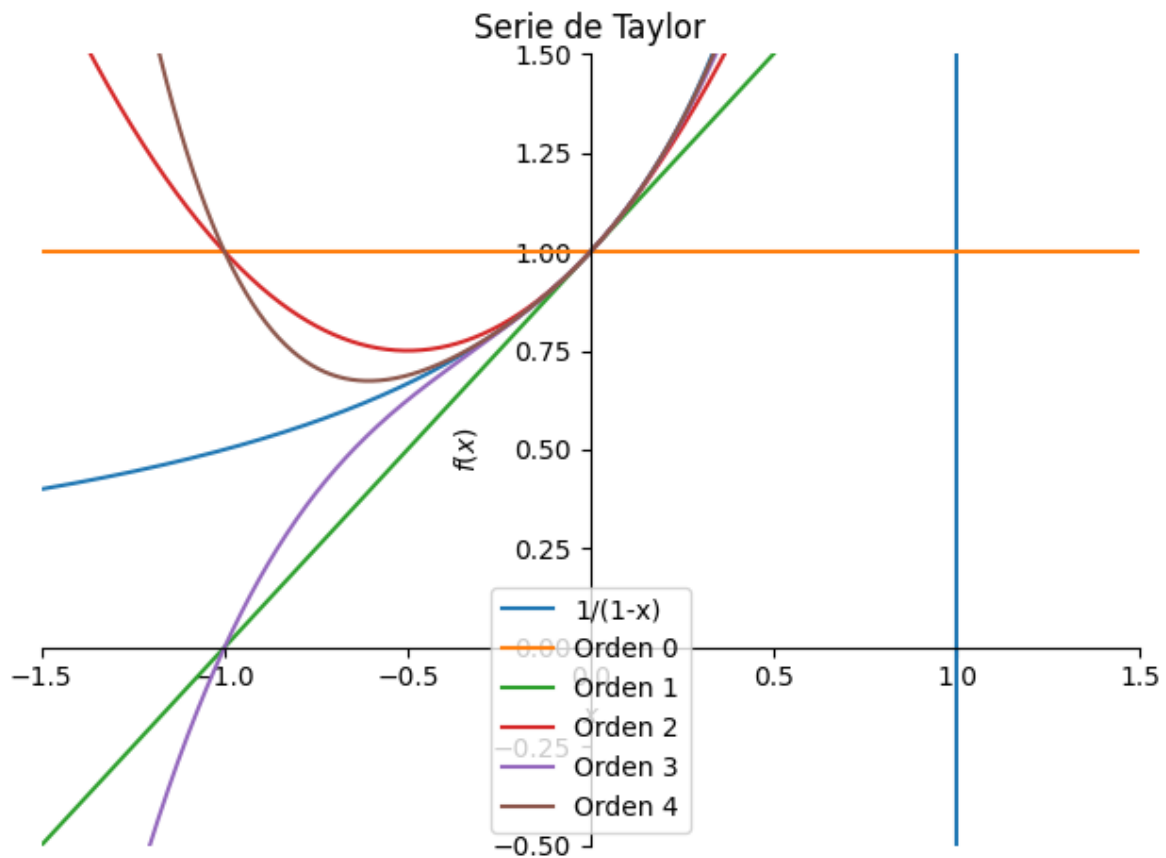
p_plot.show()

```

```

Orden 0: 1
Orden 1: x + 1
Orden 2: x**2 + x + 1
Orden 3: x**3 + x**2 + x + 1
Orden 4: x**4 + x**3 + x**2 + x + 1

```



```
import sympy as sp

x = sp.Symbol('x')
f = sp.ln(x)

# Orden
ordenes = [0, 1, 2, 3, 4]

# Funcion
graficas = [f]
labels = ['ln(x)']

for i in range(5):
    polinomio = sp.series(f, x, 1, n=i+1).removeO()
    print(f"Orden {i}: {polinomio}")

for n in ordenes:
```

```

p = sp.series(f, x, 1, n+1).remove0()
graficas.append(p)
labels.append(f'Orden {n}')

p_plot = sp.plot(*graficas, (x, -15, 15),
                 xlim=(-15, 15),
                 ylim=(-15, 15),
                 show=False,
                 title='Serie de Taylor',
                 legend=True)

# Etiqueta
for i, label in enumerate(labels):
    p_plot[i].label = label

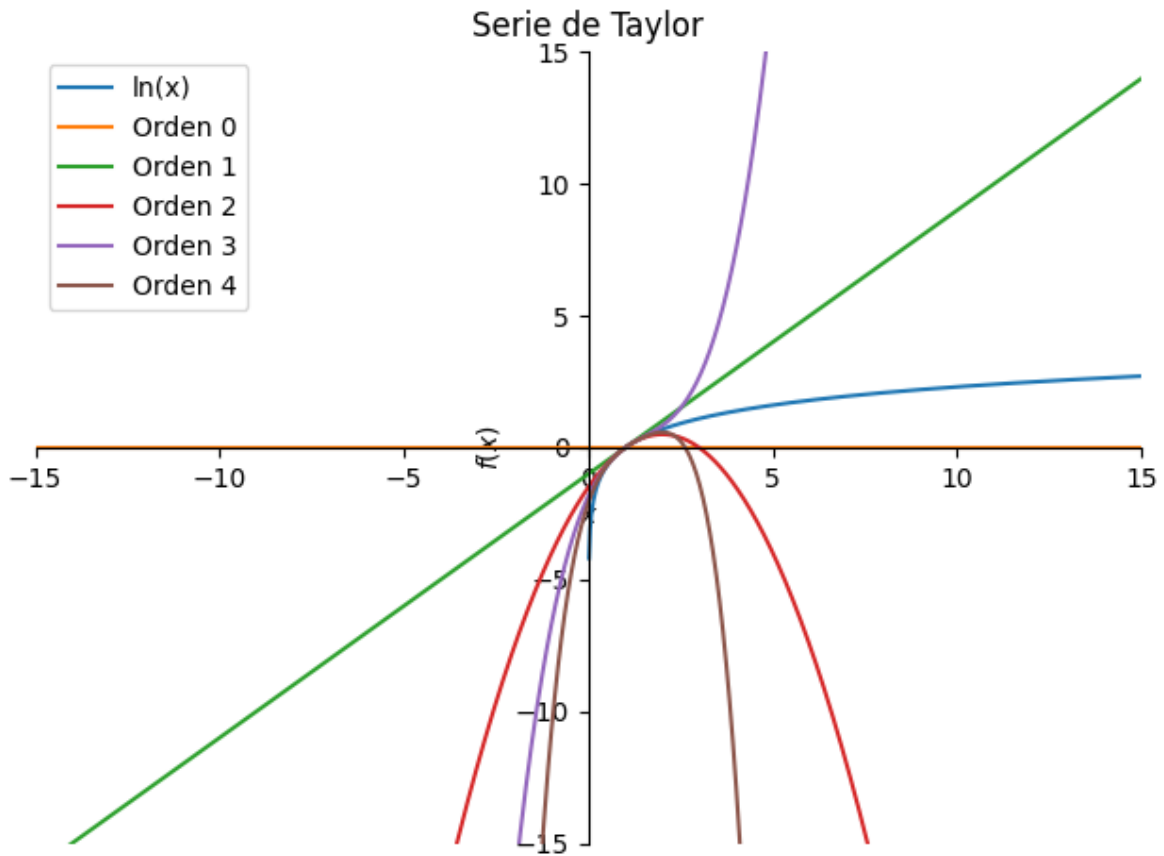
p_plot.show()

```

```

Orden 0: 0
Orden 1: x - 1
Orden 2: x - (x - 1)**2/2 - 1
Orden 3: x + (x - 1)**3/3 - (x - 1)**2/2 - 1
Orden 4: x - (x - 1)**4/4 + (x - 1)**3/3 - (x - 1)**2/2 - 1

```



Encuentre el polinomio de Lagrange para los siguientes datos y grafique:

```
import sympy as sp

x = sp.Symbol('x')

x_datos = [0, 30, 60, 90]
y_datos = [0, 0.5, float((sp.sqrt(3)/2).evalf()), 1]

# Polinomio de Lagrange
polinomio = sp.interpolate(list(zip(x_datos, y_datos)), x)

print(polinomio)

graficas = [polinomio]
labels = [f"Polinomio de Lagrange"]
```



```

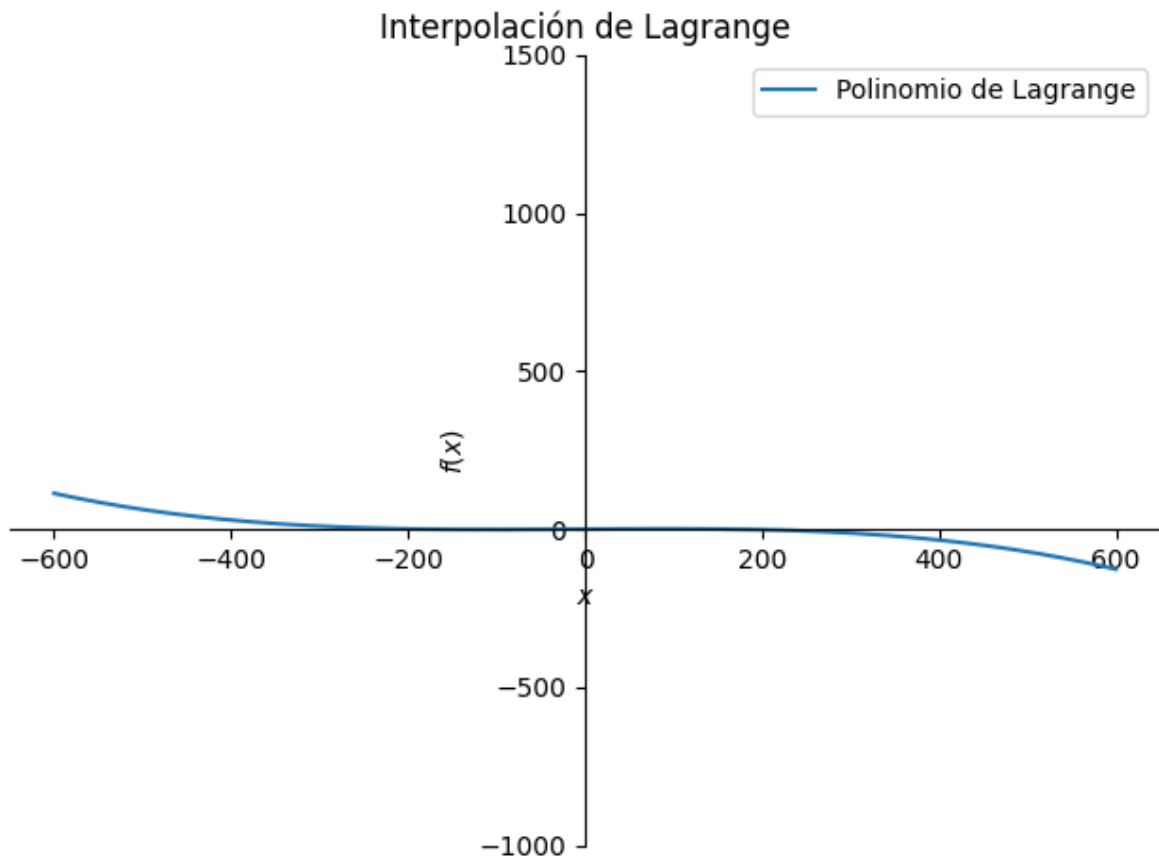
p_plot = sp.plot(
    *graficas,
    (x, -600, 600),
    xlim=(-650, 650),
    ylim=(-1000, 1500),
    show=False,
    title="Interpolación de Lagrange",
    legend=True
)

# Etiquetas
for i, label in enumerate(labels):
    p_plot[i].label = label

p_plot.show()

```

$-6.05408712057503e-7x^3 - 1.99435471456922e-5x^2 + 0.0178098409218892x$



```
import sympy as sp

x = sp.Symbol('x')

x_datos = [1, 2, 3]
y_datos = [1, 2, 2]

# Polinomio de Lagrange
polinomio = sp.interpolate(list(zip(x_datos, y_datos)), x)

print(polinomio)

graficas = [polinomio]
labels = [f"Polinomio de Lagrange"]

p_plot = sp.plot(
    *graficas,
```

```

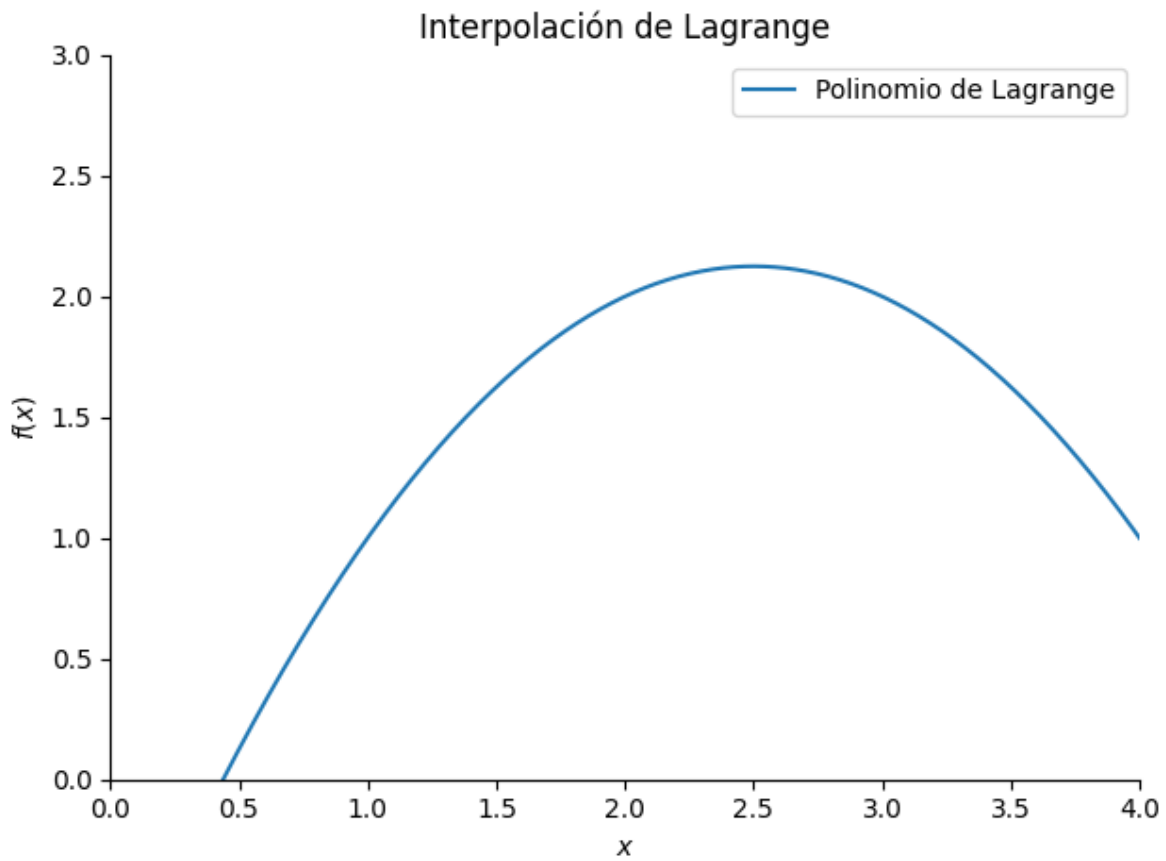
(x, 0, 4),
xlim=(0, 4),
ylim=(0, 3),
show=False,
title="Interpolación de Lagrange",
legend=True
)

# Etiquetas
for i, label in enumerate(labels):
    p_plot[i].label = label

p_plot.show()

```

$-x^2/2 + 5x/2 - 1$



```

import sympy as sp

x = sp.Symbol('x')

x_datos = [-2, 1, 3, 7]
y_datos = [5, 7, 11, 34]

# Polinomio de Lagrange
polinomio = sp.interpolate(list(zip(x_datos, y_datos)), x)

print(polinomio)

graficas = [polinomio]
labels = [f"Polinomio de Lagrange"]

p_plot = sp.plot(
    *graficas,
    (x, -30, 30),
    xlim=(-35, 35),
    ylim=(-300, 500),
    show=False,
    title="Interpolación de Lagrange",
    legend=True
)

# Etiquetas
for i, label in enumerate(labels):
    p_plot[i].label = label

p_plot.show()

```

$$43x^3/1080 + 101x^2/540 + 793x/1080 + 1087/180$$

